MIT 8103 Advanced Database Systems

E-commerce Platform - Joshua Kolo - Student ID 301905772

Case study

My case study project is centred around an E-commerce Platform. The database supports customers placing and reviewing ord
Figure 1 presents the ER model for customers, products, orders, order items, and reviews.

Portfolio 1: Database Design and Modelling

The relational design uses four core tables with primary keys, foreign keys, and checks that enforce business rules. A customer
The design also shows the relationships between customer, product, order, order item, and review entities. This keeps the data

Portfolio 2: Query Processing and Optimisation

I ran three SQL queries against the sample records. The join returned two rows and included ava@example.test with MUG-001
This exercise highlighted the value of efficient joins, selective filters, and query planning when working with larger datasets.

Portfolio 3: Transactions and Concurrency

For this task, I used a temporary counter on orders to test COMMIT, ROLLBACK, and READ COMMITTED isolation. The uncor
The exercise reinforced how transaction boundaries protect consistency while still allowing controlled concurrent access.

Portfolio 4: NoSQL and Advanced Data Models

I used order_events as the event-document collection. MongoDB inserted four sample events and returned the expected results
This demonstrated how document-based models support flexible event storage while maintaining validation rules.

Portfolio 5: Distributed and Cloud Database Exercise

Three MongoDB replica members were proposed across separate failure zones. This covered elections, majority acknowledgem
The design demonstrates how durability and availability can be balanced in distributed systems.

Individual reflection

I organized the project database by keeping customers, products, orders, and order items in separate tables. This made the dat
I also experienced a two-session concurrency issue where both sessions read the same data before either updated it. This show
Additionally, checking the execution plan alongside the results taught me that different database strategies can produce the sam

References

PostgreSQL constraints, PostgreSQL EXPLAIN, PostgreSQL isolation, PostgreSQL row locks, MongoDB data modelling, Mong